# Simultaneous equations

$x+y=1$
$x-y=0$

SIMULTANEOUS EQUATIONS ARE PAIRS OF EQUATIONS WITH THE SAME UNKNOWN VARIABLES, THAT ARE SOLVED TOGETHER.

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❮ 177–179 Formulas

## Solving simultaneous equations

Simultaneous equations are pairs of equations that contain the same variables and are solved together. There are three ways to solve a pair of simultaneous equations: elimination, substitution, and by graph; they all give the same answer.

both equations contain the variable x

both equations contain the variable y

$$3x - 5y = 4$$
$$4x + 5y = 17$$

◁ **A pair of equations** These simultaneous equations both contain the unknown variables x and y.

## Solving by elimination

Make the x or y terms the same for both equations, then add or subtract them to eliminate that variable. The resulting equation finds the value of one variable, which is then used to find the other.

▷ **Equation pair** Solve this pair of simultaneous equations using the elimination method.

$$10x + 3y = 2$$
$$2x + 2y = 6$$

**Multiply or divide** one of the equations to make one variable the same as in the other equation. Here, the second equation is multiplied by 5 to make the x terms the same.

the second equation is multiplied by 5

$$10x + 3y = 2$$

first equation stays as it is

$$2x + 2y = 6 \xrightarrow{\times 5} 10x + 10y = 30$$

this is the second equation

the x term is now the same as in the first equation

the second equation is multiplied by 5, so both equations now have the same value of x (10x)

**Then add or subtract** each set of terms in the second equation from or to each set in the first, to remove the matching terms. The new equation can then be solved. Here, the second equation is subtracted from the first, and the remaining variables are rearranged to isolate y.

this will cancel out the x terms

$$10x - 10x + 3y - 10y = 2 - 30$$

subtract the numerical terms from each other as well as the unknown terms

the x terms have been eliminated as 10x – 10x = 0

$$-7y = -28$$

this side is divided by –7 to isolate y

$$y = \frac{-28}{-7}$$

this side must also be divided by –7

$$y = 4$$

this gives the value of y

**Choose one of** the two original equations—it does not matter which—and put in the value for y that has just been found. This eliminates the y variable from the equation, leaving only the x variable. Rearranging the equation means that it can be solved, and the value of the x can be found.

$$2x + 2y = 6$$

the second equation has been chosen

$$2x + (2 \times 4) = 6$$

it is already known that y = 4 so 2y = 8

$$2x + 8 = 6$$

2 × 4 = 8

subtracting 8 from this side to isolate 2x

$$2x = -2$$

subtract 8 from this side: 6 – 8 = –2

divide this side by 2 to isolate x

$$x = \frac{-2}{2}$$

this side must also be divided by 2

$$x = -1$$

this is the value of x

**Both unknown variables** have now been found—these are the solutions to the original pair of equations.

$$x = -1 \qquad y = 4$$